

Sutirtha Ghosh

Portfolio Website

LinkedIn

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EDUCATION

- Techno Main Salt Lake** Kolkata, India
Bachelor of Technology - Electronics and Computer Science; CGPA: 8.15
2023 - 2027
- Bagnan High School** West Bengal, India
Higher Secondary Education (Class XII)
2022
- Bagnan High School** West Bengal, India
Secondary Education (Class X)
2020

SKILLS SUMMARY

- Programming Languages:** C , Python , Java
- Web Technologies:** HTML , CSS , JavaScript
- Database:** SQL
- Computer Science Fundamentals:** Data Structures and Algorithms, Object-Oriented Programming (OOP) , DBMS, Software Development Life Cycle (SDLC)
- Developer Tools:** Jupyter Notebook, VS Code, Git & GitHub
- Frameworks & Libraries:** Numpy , Scikit-learn, Pandas, PyTorch , DCGAN

EXPERIENCE

- ROADS TO INDIA** West Bengal, India
Data Analyst Intern
Jan 2025 - Jul 2025
 - Data Analysis:** Analyzed and processed 5,000+ customer and operational records using Python and SQL
 - Reporting:** Created dashboards and structured reports that improved internal data tracking efficiency by 25%
 - Data Management:** Performed data cleaning, validation, and visualization to support business decision-making
 - Collaboration:** Worked with team members to complete reporting and analytical tasks within deadlines

PROJECTS

- AI-Based Photonic Power Divider Design using DCGAN and FDTD Simulation:**
 - Developed an AI-driven inverse design framework for silicon-air photonic crystal power dividers operating at 1550 nm wavelength
 - Combined DCGANs with FDTD electromagnetic simulations based on Maxwell's equations for optical performance optimization
 - Built a hierarchical workflow with iterative GAN training and fabrication-ready 200×200 photonic layouts to improve transmission efficiency and power splitting
- Amazon E-commerce Analytics Dashboard (Python, SQL, Power BI):**
 - Developed an end-to-end analytics dashboard for Amazon product data using Python, MySQL, and Power BI
 - Performed data cleaning, exploratory data analysis, feature engineering, and SQL-based business analysis
 - Built interactive dashboards to visualize revenue, profitability, ratings, discounts, and category-wise performance
 - Used Pandas, Matplotlib, Seaborn, SQL queries, and Power BI visualizations for business insight generation

CERTIFICATIONS

- NPTEL - Object Oriented Programming:** Successfully completed certification in OOP concepts and programming fundamentals.
- NPTEL - Artificial Intelligence: Concepts and Techniques:** Learned core AI concepts, intelligent systems, and problem-solving techniques.
- NPTEL - Introduction to Machine Learning:** Completed coursework on machine learning algorithms, preprocessing, and model evaluation.

LANGUAGES

- Languages Known:** English (Professional), Bengali (Native), Hindi (Conversational)

HOBBIES

- Interests:** Problem Solving, Learning Emerging Technologies, Reading Tech Blogs, Photography